

M003 Extension 2 — English Worksheet (Intermediate)

Topic: 3D Mob Heads (x, y, z) · **Course:** 3D Coordinates · **Level:** Intermediate (Extension, after Week 3)
· **Time:** about 40 minutes

Build a **Creeper head** in 3D. Every block needs **three** numbers: **(x, y, z)**. x is **across**, y is **up**, z is **deeper** (forward). The head stands in front of you in the world. Walk around it, then copy it block by block. You build the **front face** first, then repeat it to add depth.

● Stand on your **home spot** (red block) every run. Move your feet, move your head.

1 · Predict 🧠

Read each set. Do **not** run it. Write a full sentence for each.

```
place green block at (3, 5, 1)
place green block at (3, 5, 2)
place green block at (3, 5, 3)
```

Only z changes. What happens to the line?


```
place green block at (4, 5, 1)
place green block at (4, 6, 1)
place green block at (4, 7, 1)
```

Only y changes. What happens to the line?


```
place green block at (3, 5, 1)
place black block at (3, 5, 1)
```

Same (x, y, z). What do you see, and why?

2 · Spot the Bug 🐛

Fix each one. Then say why it was wrong.

Bug A — This block should sit one step **deeper**, at (3, 5, 2). It only has two numbers, so it lands on the front.

```
place green block at (3, 5)
```

Write the fixed code:

Why was it wrong? Why does your fix work? (1–2 sentences)

Bug B — The eye should be **up high and one step deep**: (4, 7, 2). The last two numbers are swapped.

```
place black block at (4, 2, 7)
```

Write the fixed code:

Why was it wrong? Why does your fix work? (1–2 sentences)

3 · Build It 📷

Look at the  creeper head in the world. Walk around it. Then build **one full head** on your home spot, with a face.

Front face (keep $z = 1$): fill the green square, then add the creeper face — black eyes and a black mouth on the front.

Add depth: build the same front face again at $z = 2$, then $z = 3$, until your head is as deep as the real one.

Face on front only: at $z = 2$ and $z = 3$, place plain green where the eyes and mouth were.

💡 Tip: the eyes and mouth only show on the **front**. The sides and back are plain green.

Send a photo or video. Then finish these 4 sentence starters.

I built the ...

The lowest front block is at $(x = \dots, y = \dots, z = \dots)$.

To go up I changed the ... number; to go deeper I changed the ... number.

My head is ... blocks deep, and the face is on the ... side only.

4 · Explain It 🎥

Film yourself showing your head. Use these words: **x, y, z, across, up, deeper, home spot**.

1. Show your home spot. Why stand on it every time?
2. Walk around your head — show the front, a side, and the back.
3. Point at one block: "this one is at x ..., y ..., z ...".
4. Show where the face is, and explain why it is on the front only.

Write about your head — around 4 sentences:

Submit ✓

Send this worksheet + your walkthrough video to teacher on KakaoTalk.